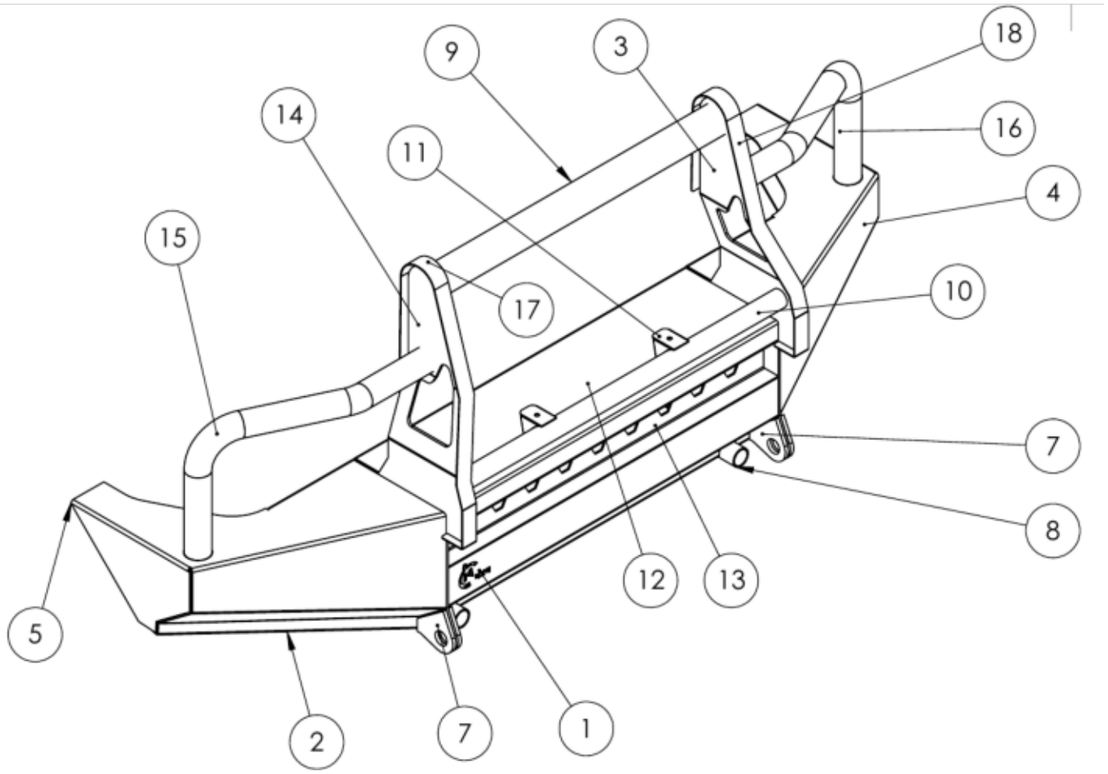
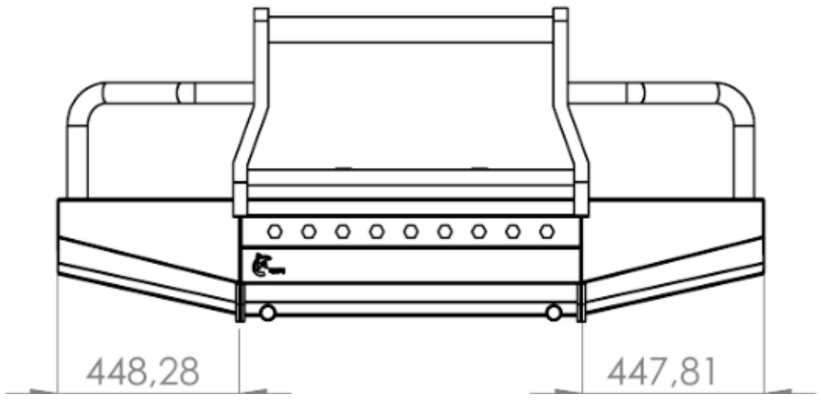
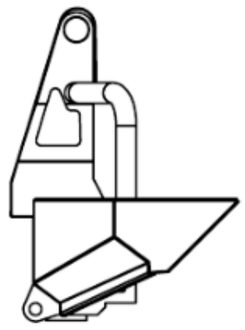
ASSEMBLY - item balloons per source BOM (see table)



FRONT / SIDE - overall 1741 mm; centre 835 mm; ends 448 mm





BILL OF MATERIALS / CUT LIST

Item	VA part no.	Description	Thk	Qty	Flat (mm)	kg ea
1	VA-BB-01	Centre panel	3	1	835 × 209	4.05
2	VA-BB-02	End panel (item 2)	3	1	688 × 687	6.32
3	VA-BB-03	Upright RH	5	1	306 × 769	4.66
4	VA-BB-04	End panel, opposite hand (item 4)	3	1	688 × 687	6.32
5	VA-BB-05	End plate	3	2	269 × 78	0.50
6	VA-BB-06	Bracket 1 (chassis bracket)	5	2	184 × 258	1.82
7	VA-BB-07	Recovery lug	8	4	91 × 92	0.29
8	VA-BB-08	Pipe lift (tube loop)	tube	2	TBC from model	-
9	VA-BB-09	Pipe centre (tube hoop)	tube	1	TBC from model	-
10	VA-BB-10	Pipe spot (light bar tube)	tube	1	TBC from model	-
11	VA-BB-11	Spot bracket (light mount)	2	2	60 × 89	0.06
12	VA-BB-12	Centre panel P2 (top plate)	3	1	835 × 289	5.73
13	VA-BB-13	Centre panel P3 (lower channel)	3	1	835 × 153	2.87
14	VA-BB-14	Upright LH	5	1	306 × 769	4.66
15	VA-BB-15	Tube 50 LH	tube	1	TBC from model	-
16	VA-BB-16	Tube 50 RH	tube	1	TBC from model	-
17	VA-BB-17	Face 1 (RH trim strip)	2	1	860 × 87	0.48
18	VA-BB-18	Face 1 LH (trim strip)	2	1	860 × 87	0.48

NEST GROUPS BY THICKNESS (laser)

2 mm 304 SS: 11 ×2, 17 ×1, 18 ×1 | net 0.07 m2 | 1.1 kg

3 mm 304 SS: 01 ×1, 02 ×1, 04 ×1, 05 ×2, 12 ×1, 13 ×1 | net 1.11 m2 | 26.3 kg

5 mm 304 SS: 03 ×1, 06 ×2, 14 ×1 | net 0.33 m2 | 13.0 kg

8 mm 304 SS: 07 ×4 | net 0.02 m2 | 1.2 kg

PROCESS NOTES

1. Laser cut all plate 1:1 from DXF (layer CUT). Layers BEND_UP / BEND_DOWN are bend-line centres - do not cut.
2. Layer ETCH = low-power surface mark (wordmark on centre panel). Layer INFO = reference text only.
3. Flats are developed at K = 0.46 (air bend, 304). 304 springs back 2-3 deg more than mild steel - over-bend and check first-off against the bend table.
4. Laser cut with nitrogen (oxide-free edge). Deburr; break sharp corners 0.5 mm. Stainless-only tooling, films and brushes - no carbon-steel contamination.
5. Bend direction is viewed from the flat-pattern face shown on each sheet (UP = flange toward viewer).
6. Tube items 8/9/10/15/16 are not in this plate pack - lengths and bend data to come from the 3D model.
7. Items 17/18 (face strips): source drawing states "NO BENDING" yet carries bend notes - confirm with design before forming.
8. Weld sequence: tack brackets (06) to centre (01) and P2/P3 (12/13); fit uprights (03/14) and lugs (07); add ends (02/04) and end plates (05) last.
9. Several source bend radii are below 1T (red warnings on sheets 05, 06, 11, 03/14, 17/18). Workable in 304 with a sharp-nose punch; confirm on first-off - if it cracks, open the radius.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-ASM

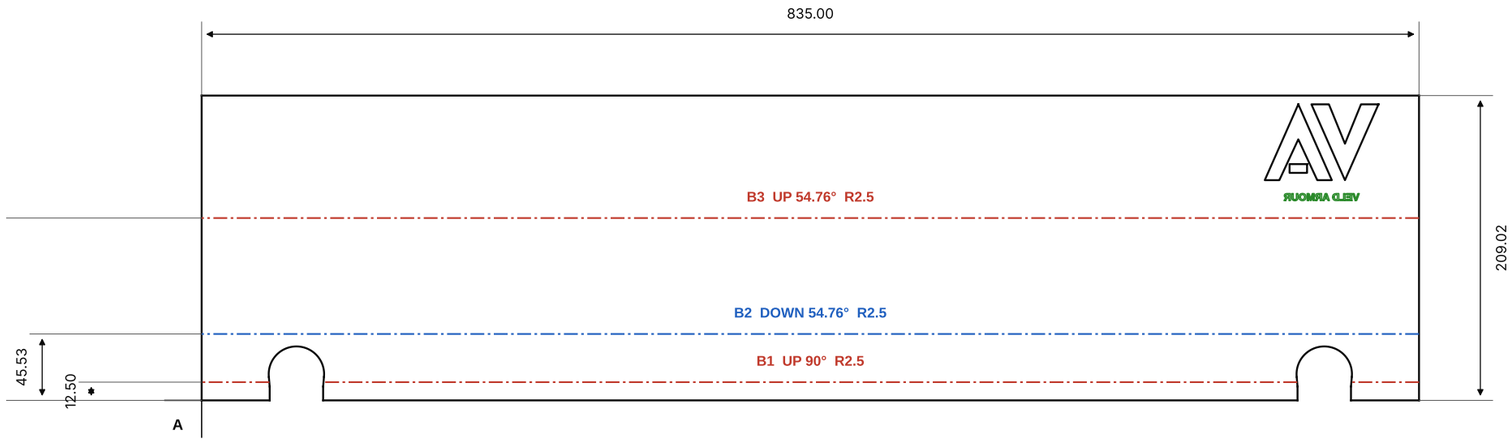
TITLE
BULL BAR - ASSEMBLY, BOM & CUT LIST

MATERIAL **304 SS 2B**	SHEET PARTS **19**	PLATE MASS **41.5 kg (excl. tube)**
OVERALL WIDTH 1741 mm	CENTRE SECTION 835 mm	END SECTIONS 448 mm each
THICKNESSES 2 / 3 / 5 / 8 mm	FITMENT TBC (per source 4×4 design)	TUBE 50 mm dia (per BOM)
WELDING TIG/MIG, ER308L, purge	FINISH Brushed, passivated	PACK VA-BB rev A

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

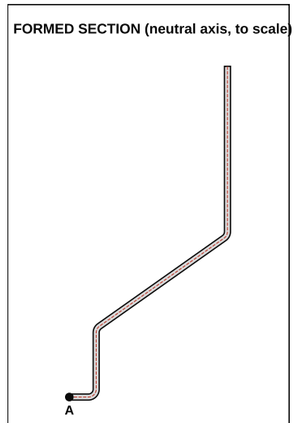


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2.5	12.50 (Y)	6.09	4.91	24
B2	DOWN	54.76°	2.5	45.53 (Y)	3.71	1.99	24
B3	UP	54.76°	2.5	125.04 (Y)	3.71	1.99	24

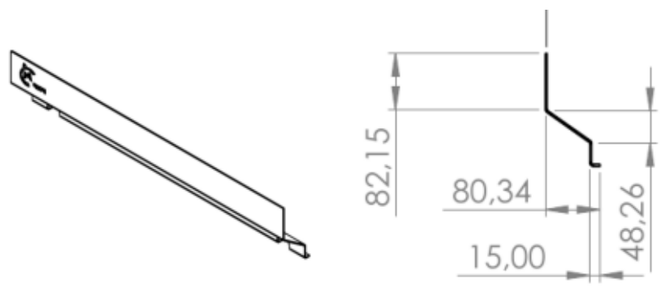
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 12.5 / 33.0 / 79.5 / 84.0 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Veld Armour monogram cut through; wordmark on ETCH layer (mirrored in flat, reads correct on outside face). (2) Half-round notches clear the upright lugs - check fit-up before final bend.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-01

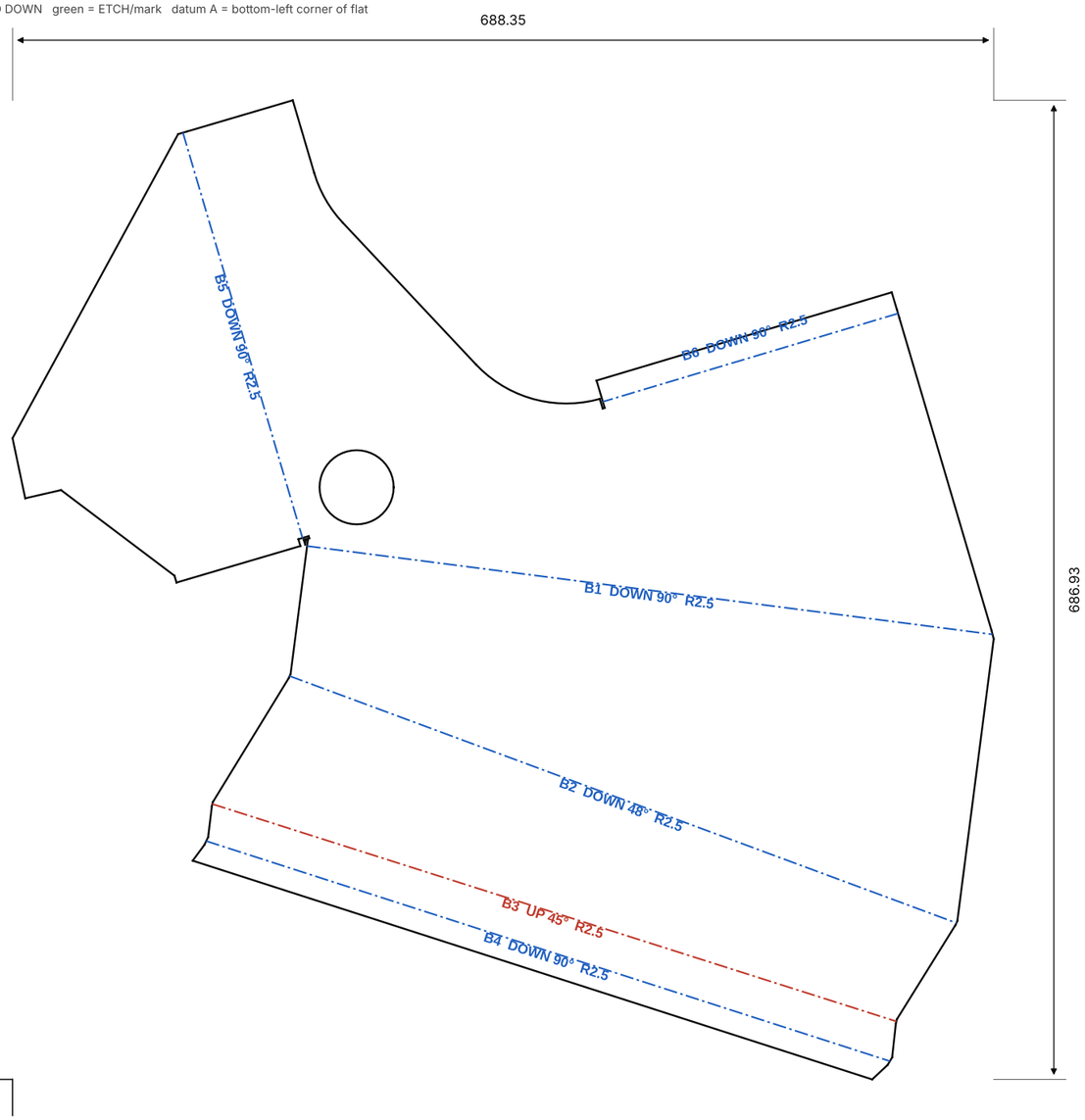
TITLE
Centre panel - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 835.0 x 209.0	NET MASS 4.05 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 3	SOURCE ITEM #1	DXF VA-BB-01_Centre_panel_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

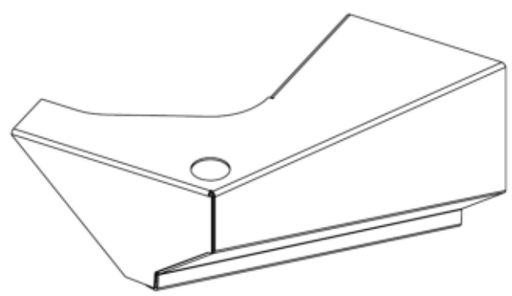
No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	2.5	see view	6.09	4.91	24
B2	DOWN	48°	2.5	see view	3.25	1.65	24
B3	UP	45°	2.5	see view	3.05	1.51	24
B4	DOWN	90°	2.5	see view	6.09	4.91	24
B5	DOWN	90°	2.5	see view	6.09	4.91	24
B6	DOWN	90°	2.5	see view	6.09	4.91	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3, B4, B5, B6 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)

Non-parallel bends - see formed view and assembly sheet for orientation.

FORMED VIEW (reference, not to scale)



NOTES: (1) Non-parallel bends: form the short 90 deg corner flanges last, after the 48/45/90 return lip. (2) Suggested sequence: B2 (48 deg) > B3 (45 deg up) > B4 (90 deg lip) > B1 (long 90) > B5, B6 (corner flanges).

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-02

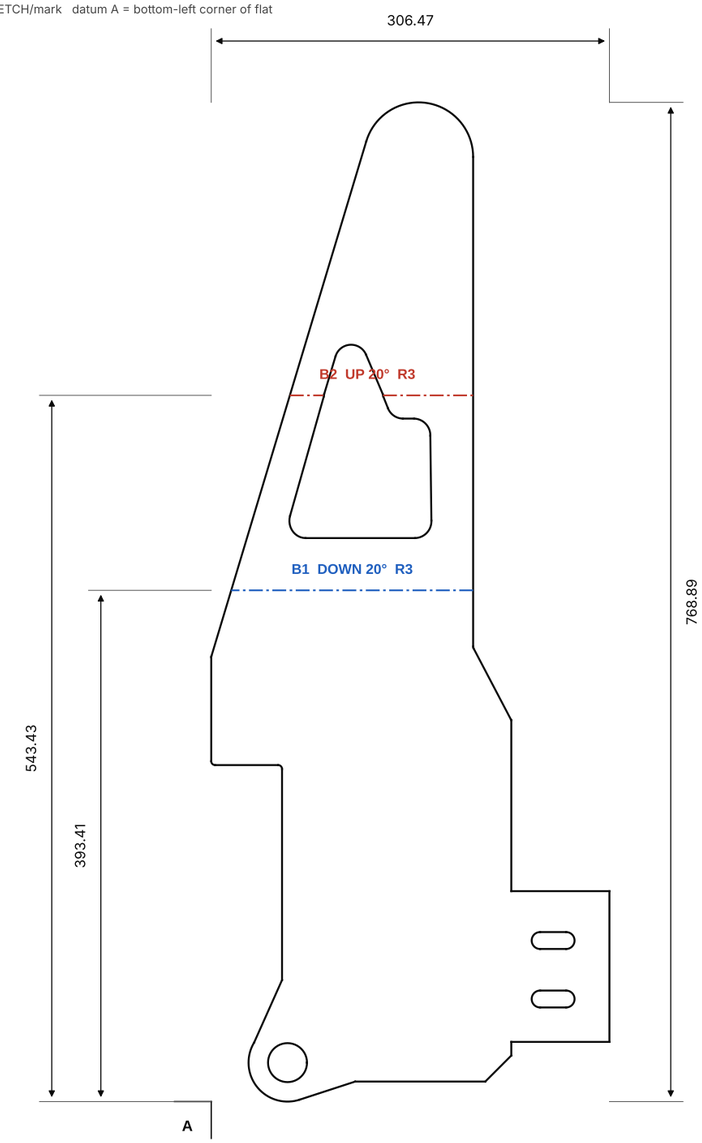
TITLE
End panel (item 2) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 688.3 × 686.9	NET MASS 6.32 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 6	SOURCE ITEM #2	DXF VA-BB-02_End_panel_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	20°	3	393.41 (Y)	1.85	0.97	40
B2	UP	20°	3	543.43 (Y)	1.85	0.97	40

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 393.4 / 150.0 / 225.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Two opposed 20 deg bends form a joggle - set back gauge from the slotted foot (datum edge A). (2) Lug (VA-BB-07) welds over the dia 30 hole; R30 boss on this part matches lug outer radius.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-03

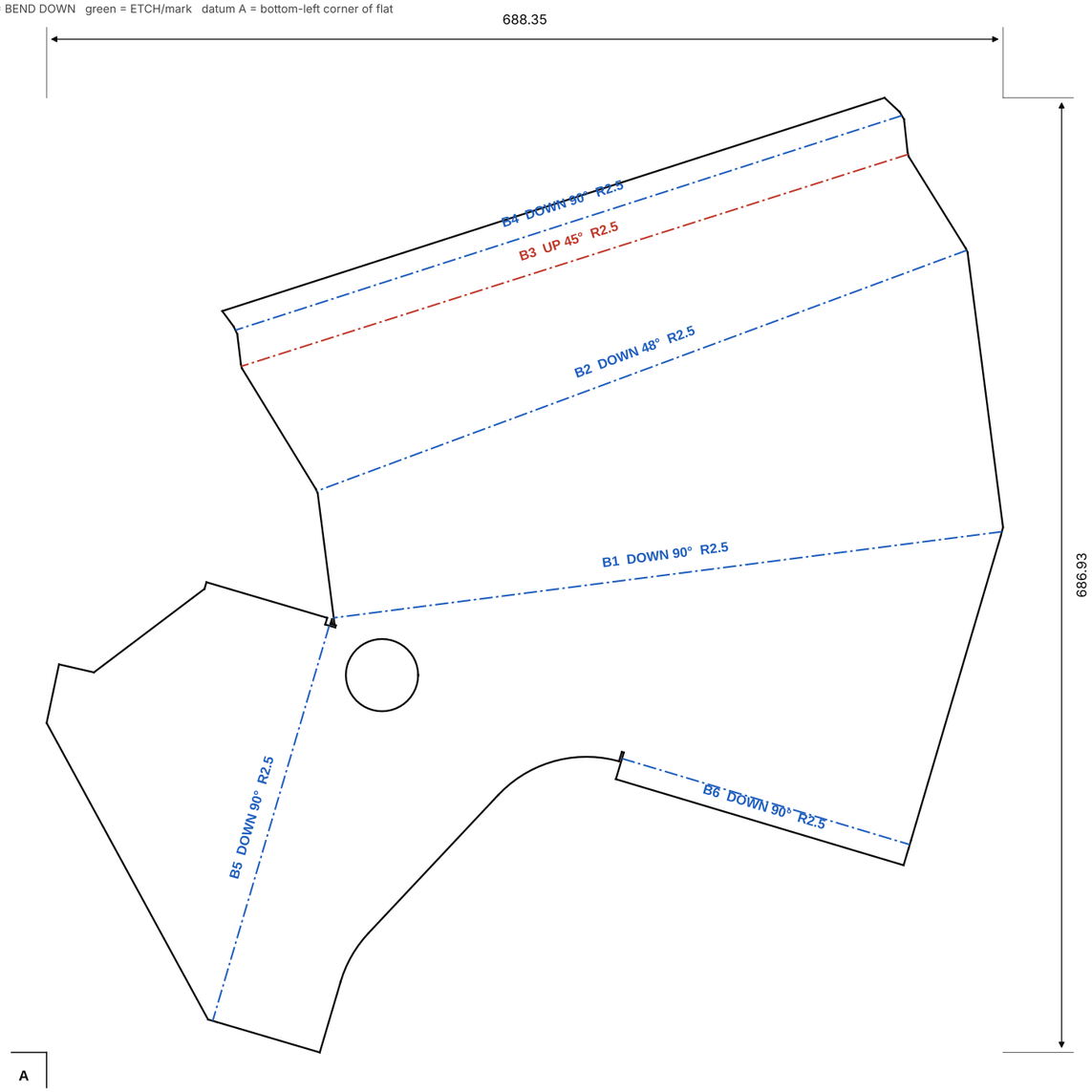
TITLE
Upright RH - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **5 mm**	QTY / ASSY **1**
FLAT SIZE 306.5 × 768.9	NET MASS 4.66 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #3	DXF VA-BB-03_Upright_RH_5mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

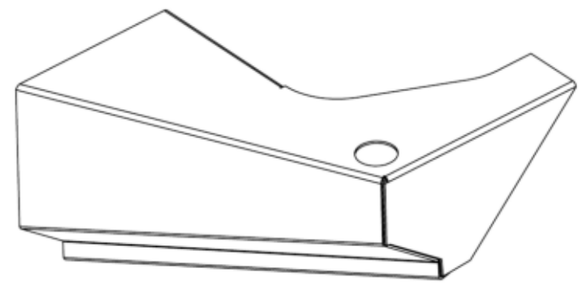
No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	2.5	see view	6.09	4.91	24
B2	DOWN	48°	2.5	see view	3.25	1.65	24
B3	UP	45°	2.5	see view	3.05	1.51	24
B4	DOWN	90°	2.5	see view	6.09	4.91	24
B5	DOWN	90°	2.5	see view	6.09	4.91	24
B6	DOWN	90°	2.5	see view	6.09	4.91	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3, B4, B5, B6 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)

Non-parallel bends - see formed view and assembly sheet for orientation.

FORMED VIEW (reference, not to scale)



NOTES: (1) Mirror of VA-BB-02. Same bend sequence as VA-BB-02.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-04

TITLE
End panel, opposite hand (item 4) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**3 mm**	**1**
FLAT SIZE	NET MASS	K-FACTOR
688.3 × 686.9	6.32 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
6	#4	VA-BB-04_End_panel,_opposite_hand_3mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:2

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	40.84°	1.5	243.32 (X)	2.05	1.30	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 243.3 / 25.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Single shallow bend; use a sharp-radius V-die (V = 6T = 18 mm).

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-05

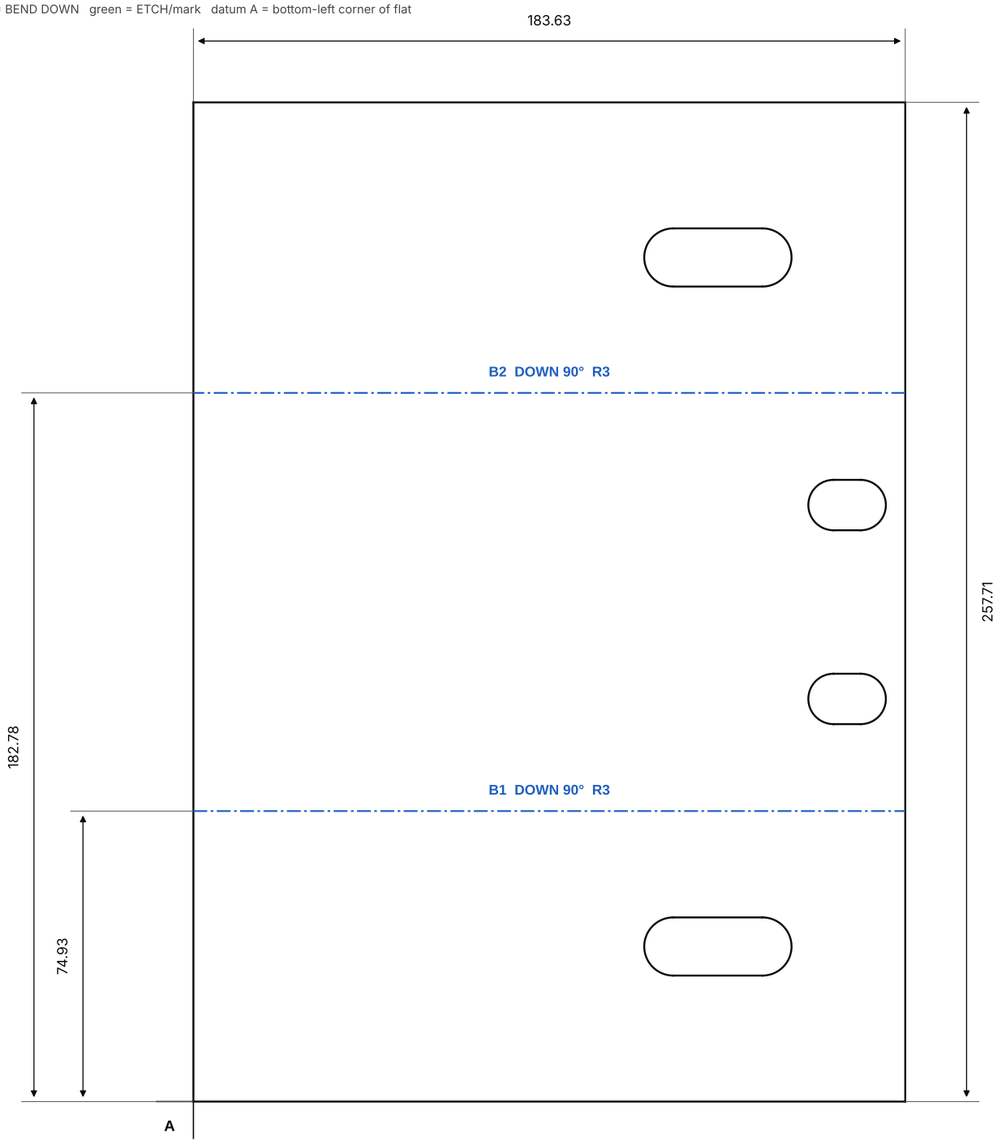
TITLE
End plate - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**3 mm**	**2**
FLAT SIZE	NET MASS	K-FACTOR
268.8 × 78.0	0.50 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
1	#5	VA-BB-05_End_plate_3mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:2

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

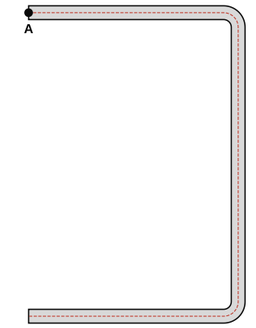


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	3	74.93 (Y)	8.33	7.67	40
B2	DOWN	90°	3	182.78 (Y)	8.33	7.67	40

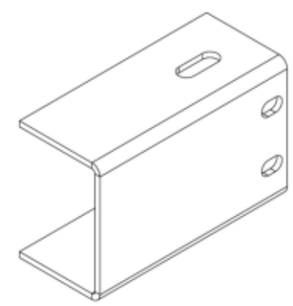
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 74.9 / 107.9 / 74.9 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) U-channel: bend both legs in the same direction. Check slot alignment to chassis rail after forming.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-06

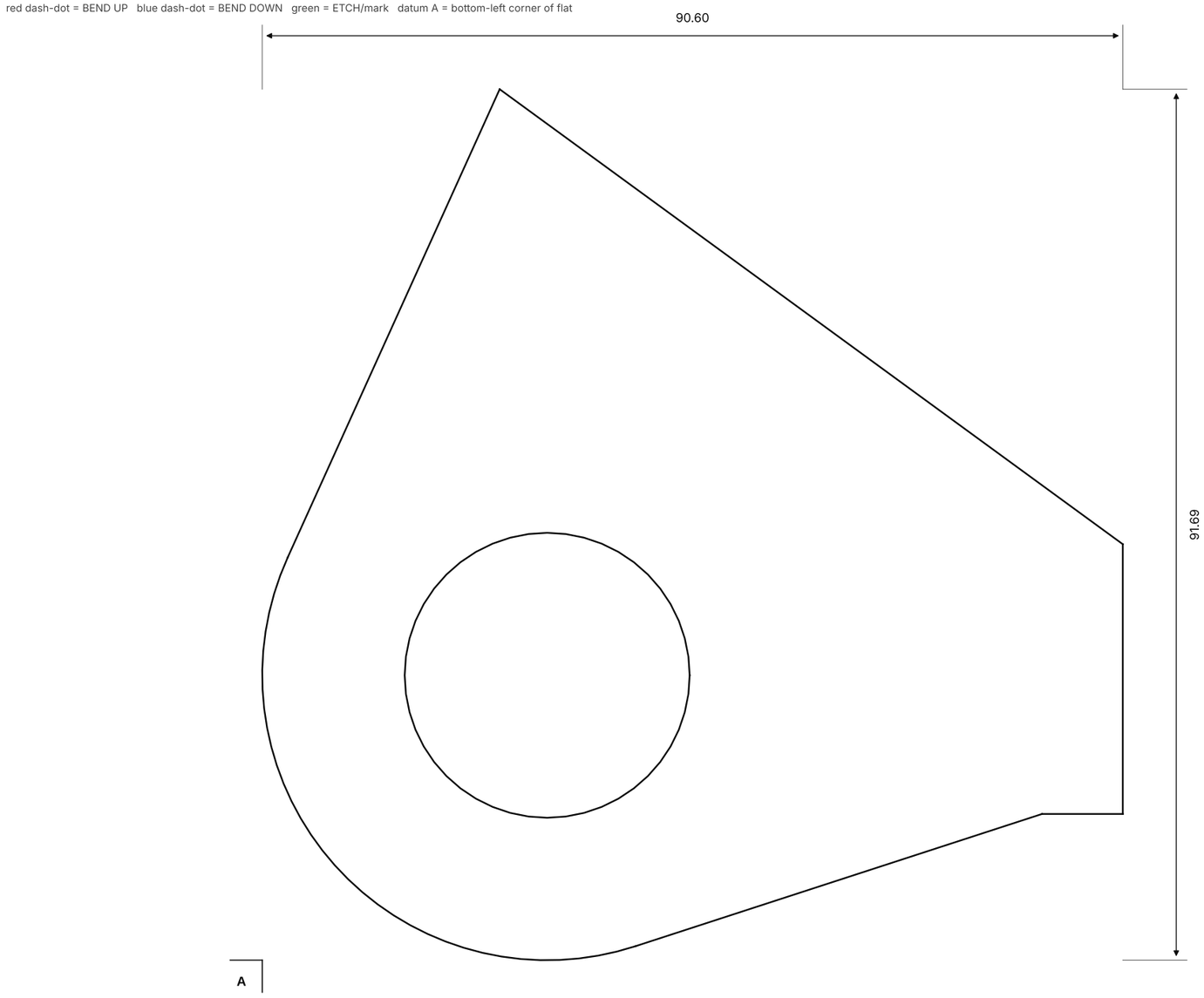
TITLE
Bracket 1 (chassis bracket) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**5 mm**	**2**
FLAT SIZE	NET MASS	K-FACTOR
183.6 × 257.7	1.82 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
2	#6	VA-BB-06_Bracket_1_5mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:1

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

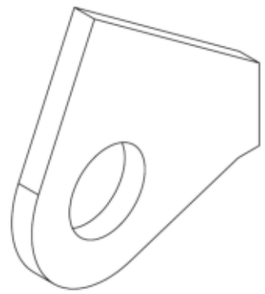
No	Dir	Angle	R in	From A	BA	BD	V-die
No bends - flat laser-cut part.							

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.

FORMED SECTION (neutral axis, to scale)

Flat part - no section.

FORMED VIEW (reference, not to scale)



NOTES: (1) Redrawn from assembly drawing sheet 8 (no flat supplied in source). Dia 30 hole, R30 boss. (2) Laser or plasma cut 8 mm; dress the hole for a 30 mm pin / shackle bow.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-07

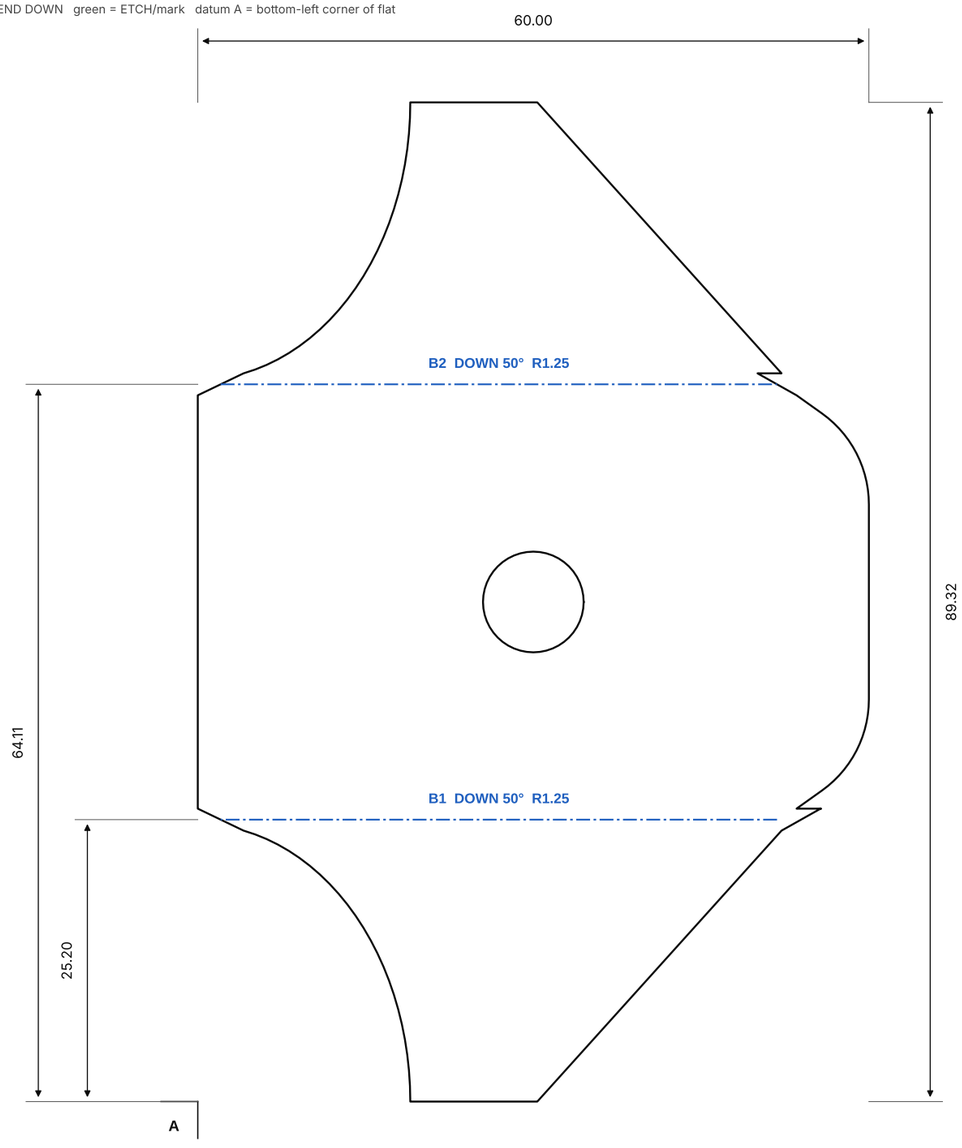
TITLE
Recovery lug - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**8 mm**	**4**
FLAT SIZE	NET MASS	K-FACTOR
90.6 × 91.7	0.29 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
0	#7	VA-BB-07_Recovery_lug_8mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:1

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

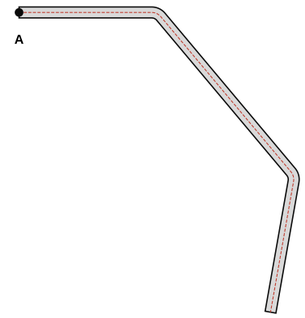


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	50°	1.25	25.20 (Y)	1.89	1.14	16
B2	DOWN	50°	1.25	64.11 (Y)	1.89	1.14	16

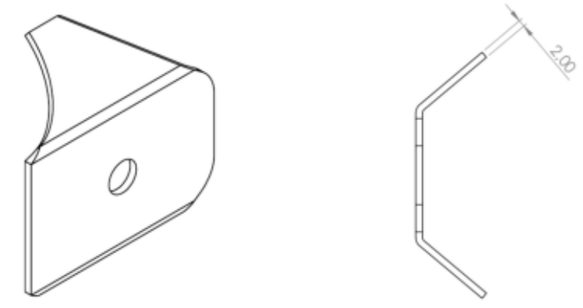
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 25.2 / 38.9 / 25.2 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Both wings bend the same way (down) to form a saddle. Dia 9 hole for M8 spot-light stud.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-11

TITLE
Spot bracket (light mount) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **2 mm**	QTY / ASSY **2**
FLAT SIZE 60.0 x 89.3	NET MASS 0.06 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #11	DXF VA-BB-11_Spot_bracket_2mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

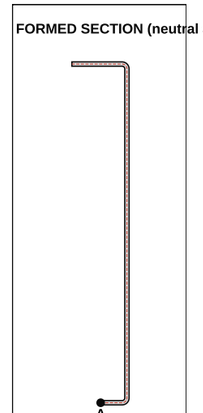


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2	17.51 (Y)	5.31	4.69	24
B2	UP	90°	2	251.18 (Y)	5.31	4.69	24

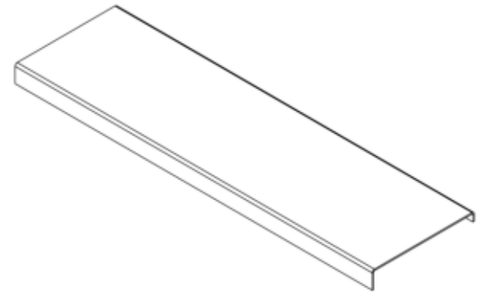
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 17.5 / 233.7 / 37.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Hat/channel section with unequal legs (17.5 and 37.5 mm). Short leg first.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-12

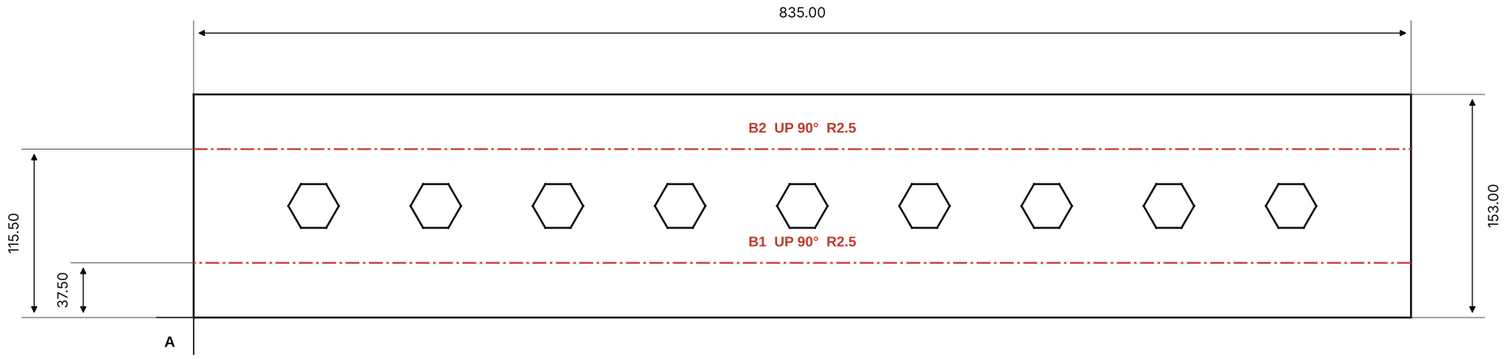
TITLE
Centre panel P2 (top plate) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 835.0 x 288.7	NET MASS 5.73 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #12	DXF VA-BB-12_Centre_panel_P2_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2.5	37.50 (Y)	6.09	4.91	24
B2	UP	90°	2.5	115.50 (Y)	6.09	4.91	24

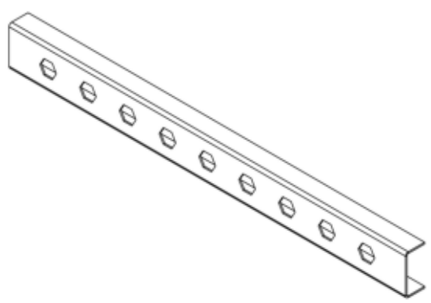
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 37.5 / 78.0 / 37.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) 9 x hex apertures (20 mm A/F) on the web - decorative/vent. Equal 37.5 mm legs.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-13

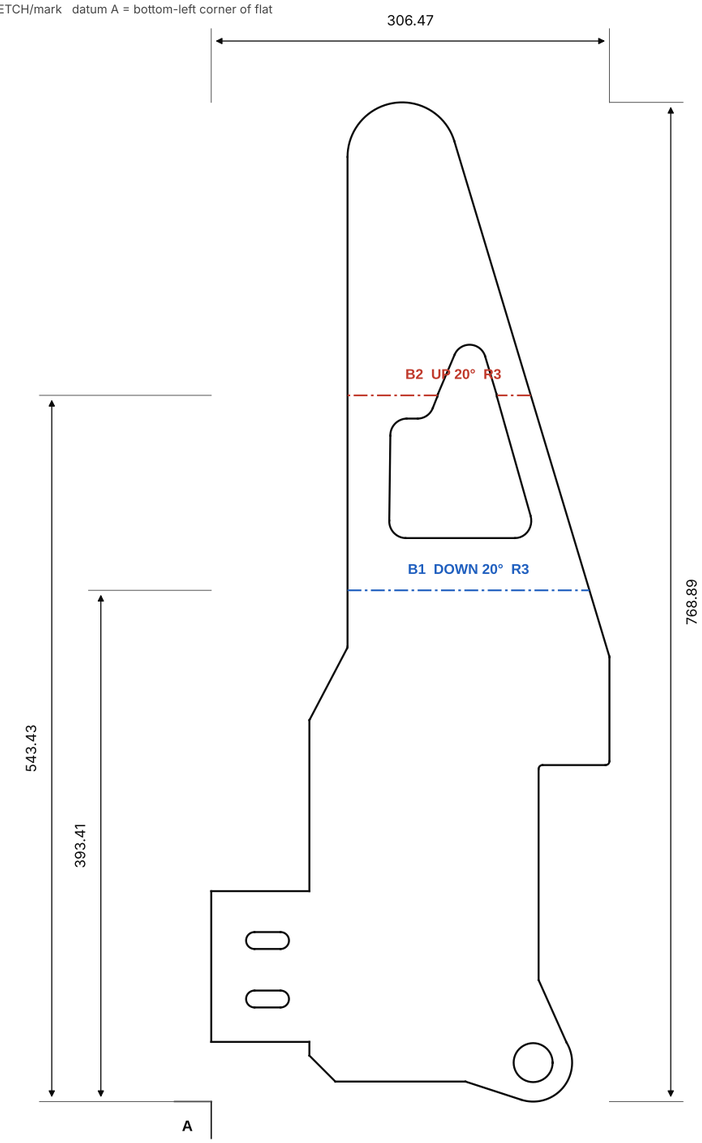
TITLE
Centre panel P3 (lower channel) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 835.0 x 153.0	NET MASS 2.87 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #13	DXF VA-BB-13_Centre_panel_P3_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

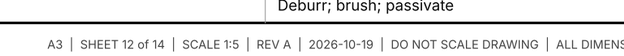


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	20°	3	393.41 (Y)	1.85	0.97	40
B2	UP	20°	3	543.43 (Y)	1.85	0.97	40

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 393.4 / 150.0 / 225.5 mm

FORMED VIEW (reference, not to scale)



opposite hand shown

NOTES: (1) Mirror of VA-BB-03 (generated). Bend directions unchanged.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-14

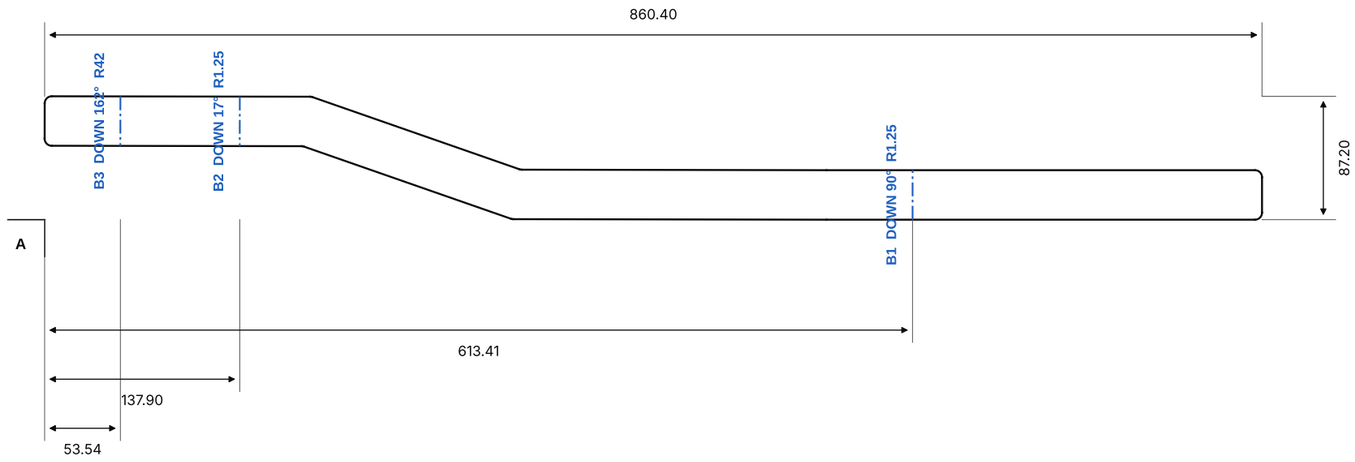
TITLE
Upright LH - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **5 mm**	QTY / ASSY **1**
FLAT SIZE 306.5 × 768.9	NET MASS 4.66 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #14	DXF VA-BB-14_Upright_LH_5mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	1.25	613.41 (X)	3.41	3.09	16
B2	DOWN	17°	1.25	137.90 (X)	0.64	0.33	16
B3	DOWN	162°	42	53.54 (X)	121.35	434.26	roll

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 53.5 / 84.4 / 475.5 / 247.0 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) SOURCE DRAWING CARRIES THE NOTE "NO BENDING" ON THIS PART while also showing three bend callouts. (2) Bend lines are supplied as drawn. CONFIRM WITH DESIGN before forming. R42 bend = roll/bump form.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-17

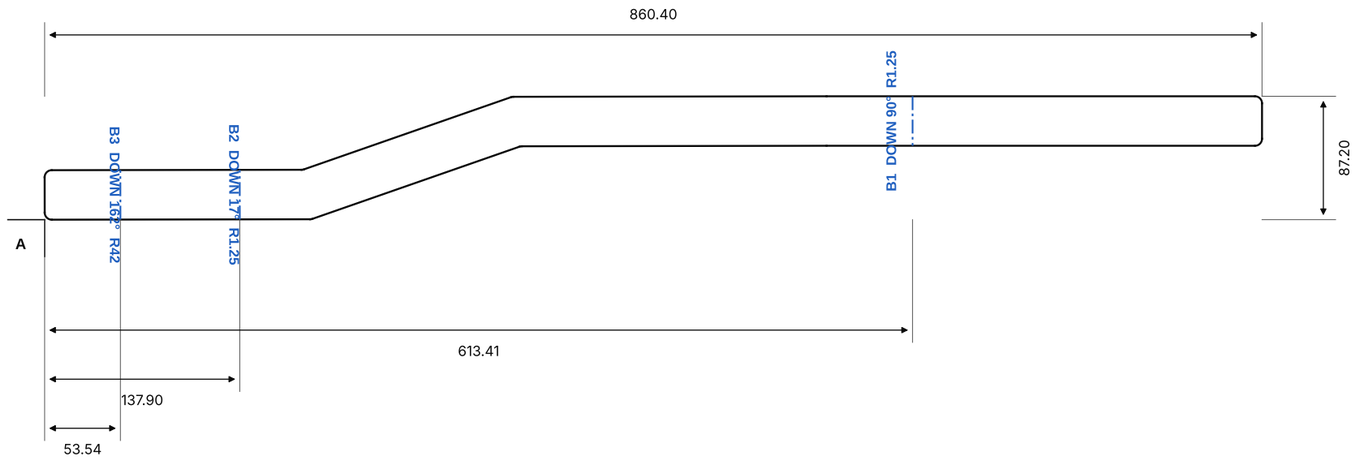
TITLE
Face 1 (RH trim strip) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **2 mm**	QTY / ASSY **1**
FLAT SIZE 860.4 × 87.2	NET MASS 0.48 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 3	SOURCE ITEM #17	DXF VA-BB-17_Face_1_2mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	1.25	613.41 (X)	3.41	3.09	16
B2	DOWN	17°	1.25	137.90 (X)	0.64	0.33	16
B3	DOWN	162°	42	53.54 (X)	121.35	434.26	roll

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 53.5 / 84.4 / 475.5 / 247.0 mm

FORMED VIEW (reference, not to scale)



opposite hand shown

NOTES: (1) Mirror of VA-BB-17 (generated). Same "NO BENDING" caveat applies - confirm before forming.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-18

TITLE
Face 1 LH (trim strip) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **2 mm**	QTY / ASSY **1**
FLAT SIZE 860.4 × 87.2	NET MASS 0.48 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 3	SOURCE ITEM #18	DXF VA-BB-18_Face_1_LH_2mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

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